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Thermally Stable Conformance Control

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Abstract

Mukhaizna field in southern Oman, characterized by heavy oil reservoirs with high viscosity and low API gravity, has relied extensively on steam flooding to improve oil mobility and recovery. However, the technique faces significant conformance challenges due to steam breakthrough and inefficient sweep, especially in thief zones. To address these issues, a thermally stable conformance control chemical was developed and field-tested across multiple phases. This solution, deployed via coiled tubing with inflatable packers, targeted specific reservoir intervals while operating under high-temperature and high-H_2S conditions.

The project was executed over four years and involved more than 100 wells, each selected based on poor injectivity profiles and pronounced thief zone behavior. The deployment method included customized coiled tubing bottomhole assemblies, temperature mitigation strategies, and zonal isolation using inflatable packers. Evaluation techniques included pre- and post-job spinner logging, surface pressure monitoring, and injectivity testing.

Key findings include a significant improvement in steam injection conformance, as evidenced by surface pressure behavior and subsurface logs. Pre- and post-treatment results demonstrated a shift from 89% of steam being absorbed by the upper zone to a balanced 46%/54% distribution between upper and lower zones. This led to a measurable decrease in lifting costs, an increase in oil recovery, and a reduction in CO_2 emissions. Furthermore, the project integrated locally manufactured raw materials, contributing to in-country value (ICV) and SME development. The outcomes validate the effectiveness of this approach and its scalability across similar thermal EOR environments. The methodology and outcomes discussed here can serve as a benchmark for future high-temperature conformance control applications.

Background

Mukhaizna field, located in southern Oman, is characterized by heavy oil reservoirs with oil densities ranging between 14 to 18°API and viscosities exceeding 1,000 cP. To reduce this high viscosity and enhance oil mobility, steam injection is employed to support production wells—a process widely known as steam flooding, one of the most established thermal enhanced oil recovery (EOR) techniques [1][2]. As described in [1] , steam management undergoes complex surveillance such as, minimizing steam: oil ratio – barrel of steam injected to produce barrel of oil-, pattern configuration of oil injector supported by steam injectors, steam breakthough early signs and identifying suitable steam injectors drilled locations. Thus, much effort has been done to ensure economic feasibility and increase steam sweep efficiency.

While steam flooding aims to significantly improved oil recovery, it presents a number of operational challenges. One of the primary concerns is steam breakthrough, which occurs due to inadequate sweep efficiency within the reservoir. This condition leads to premature steam arrival at producers, bypassed oil, inefficient displacement, and ultimately reduced oil production [1]. Poor conformance control intensifies this issue, increasing lifting costs and reducing thermal efficiency.

To address these challenges, a thermally stable conformance control chemical was engineered and deployed using coiled tubing with inflatable packers to selectively isolate non-target (non-thief) zones. In addition, a bullheading technique was introduced to treat multiple zones simultaneously, especially under high-temperature conditions exceeding 500°F. These methods aim to improve steam sweep efficiency, enhance oil recovery, reduce lifting costs, and lower CO_2 emissions. After successful pilot stage as detailed in [3], the project advanced into subsequent phases with enhancements driven by operational feedback and lessons learned.

This paper focuses on documenting the execution journey during Phases 2 and 3, beginning with job design and chemical selection, followed by intervention strategy utilizing the inflatable packer along with engineering mitigation to execute the job safely under high temperature (>500 °F) and elevated H2S condition. A pre and post injectivity results supported by logging data showcased the effectiveness of this innovative method and demonstrated steam distribution across the zones. The study concludes with key learnings from the implementation of this conformance solution in the Mukhaizna steam flood environment.

Job Design and deployment technique

The primary objective of the conformance solution was to develop a thermally stable chemical capable of selectively blocking thief zones in a controlled manner under high-temperature reservoir conditions. Multiple solutions were screened during the early stages of the project and assessed based on data provided by client, as illustrated in Fig. 1.

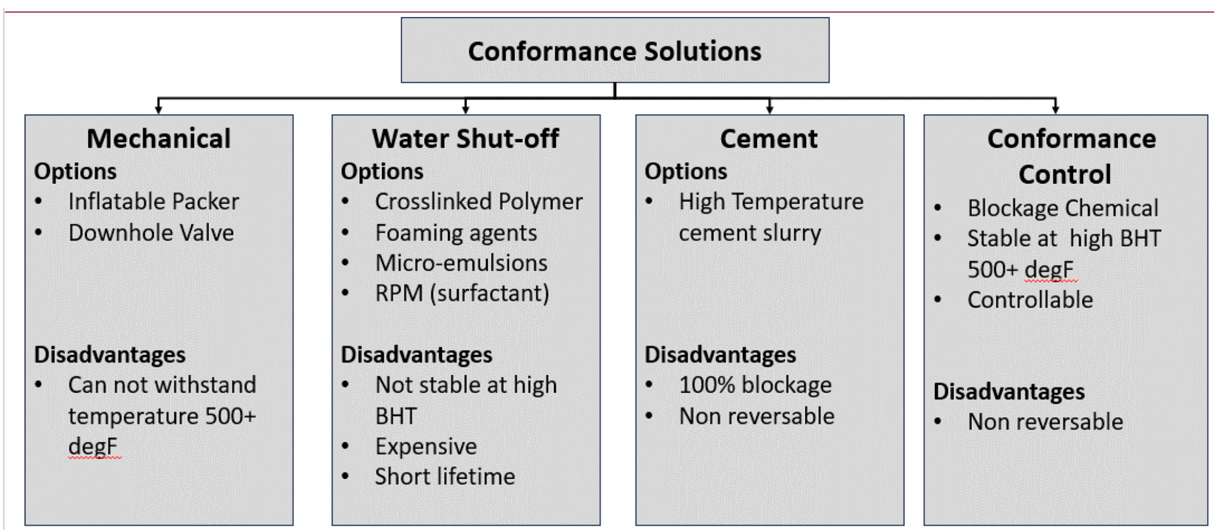

Figure 1—Solution Matrix of conformance control solutions and their limitations

In addition to the qualitative assessment in Fig. 1, a relative comparison highlights the practical differences between the solutions. Mechanical options, despite low upfront cost, lose effectiveness entirely once reservoir temperature exceeds 500 °F, making them unsuitable in this case. Water shut-off methods, although moderately effective initially, are penalized by high chemical cost and limited lifetime, resulting in higher $/bbl of treated fluid when evaluated over sustained injection periods. Cement placement provides long-term durability but creates 100% blockage, which reduces reservoir productivity and increases remedial costs in the event of future interventions. The chemical conformance system, while also non-reversible, maintains stability above 500 °F and provides selective control of thief zones, achieving higher effectiveness per unit volume placed with lower overall operational complexity. This balance of cost, durability, and effectiveness justifies its selection over mechanical and cementing approaches under high-temperature steamflood conditions.

Conformance control solution proposed, is inorganic blockage material designed to activate within pore throats, creating a plug like effect narrowing the throat radius, thus enabling effective conformance control. The reaction is sensitive to temperature, allowing it to harden progressively as the temperature increases, resulting in a stable product at high bottom hole conditions. The kinetic of reaction is controlled by activator concentration, enabling a delayed reaction until the chemical is properly in place across the zone. This delay is crucial to avoid premature setting and to ensure uniform placement within the thief zones. Prior to pilot phase implementation as described in [3], extensive lab tests have been done to evaluate the sustainability of the solution under reservoir conditions. Laboratory results demonstrated that the solution was capable of reducing permeability from 3,000 mD to 100 mD when exposed to a temperature of 500°F. This reduction in permeability was sustained over a 8 week observation period, confirming the chemical's long-term stability and effectiveness in maintaining the plug effect under high-pressure, high-temperature (HPHT) conditions. These findings validated the solution's suitability for application in harsh downhole environments, providing the necessary confidence to proceed with field-scale deployment.

Chemical placement was executed via coiled tubing (CT) with the utilization of inflatable packers. These packers were single-use and activated through a ball-drop mechanism and were strategically positioned either above or below the targeted reservoir zone, depending on the specific treatment objective. One of the key operational challenges encountered was that the bottomhole temperature (BHT) exceeded the maximum temperature rating of the selected packer. Given that the wellbore was 3.5 inches outer diameter (OD) with a minimum internal restriction of 2.4 inches, the only feasible packer that could be deployed was a 2.125-inch model. This resulted in a tight clearance of 0.275 inches. The selected packer was rated for a working temperature of 300°F and a differential pressure capacity of 5,000 psi, and was designed to expand to seal against the 3.5-inch tubing size. As will be further discussed in later sections, achieving effective mechanical separation between the targeted and non-targeted zones was crucial to the success of the conformance control treatment. To overcome the temperature limitation of the packer, the well was shut in for over three days to allow both bottomhole and surface (Christmas tree) temperatures to cool sufficiently. In parallel, and during the packer deployment, water was continuously pumped into the wellbore to further decrease the thermal load and ensure that the packer elastomers were not compromised by elevated temperatures.

This placement strategy played a key role in isolating zones that received minimal steam sweep from those acting as thief zones, thereby enabling the conformance control solution to selectively plug high-permeability channels receiving the majority of injected steam. Additionally, the inflatable packers served to protect the under-swept intervals from unintended exposure to the chemical treatment. Figures 2 and 3 illustrate the coiled tubing-based well intervention solution and the associated deployment technique, respectively.

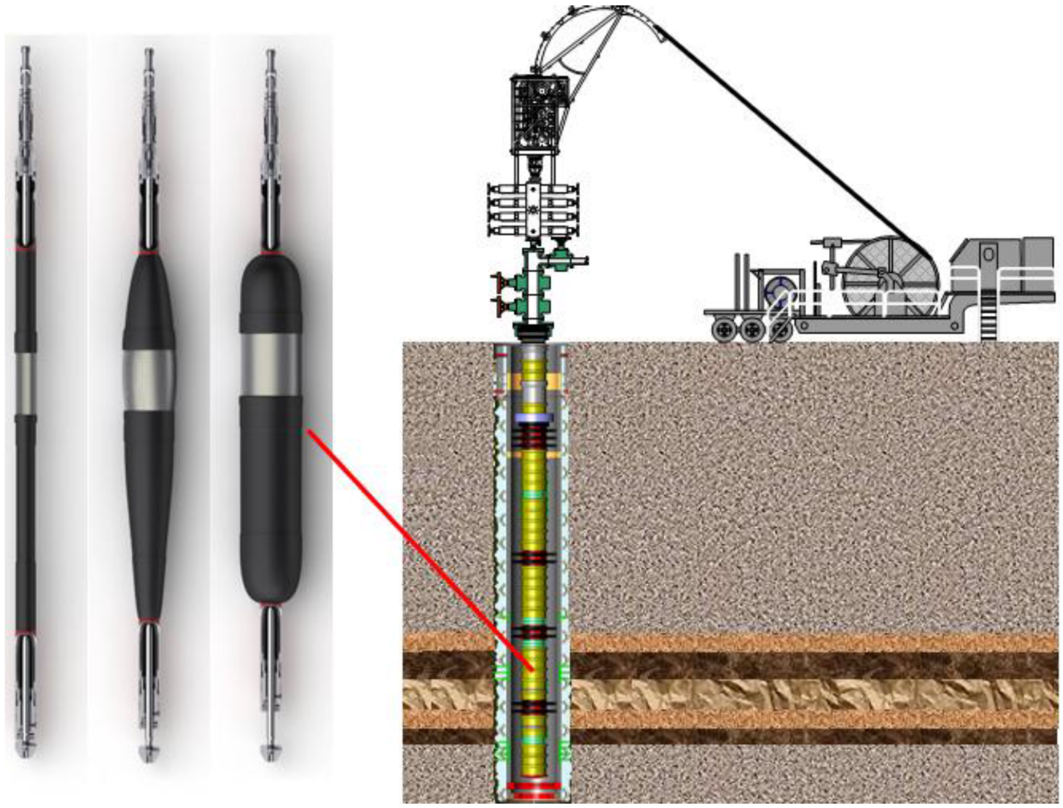

Figure 2—Coiled tubing intervention utilizing an inflatable packer for selective zonal isolation

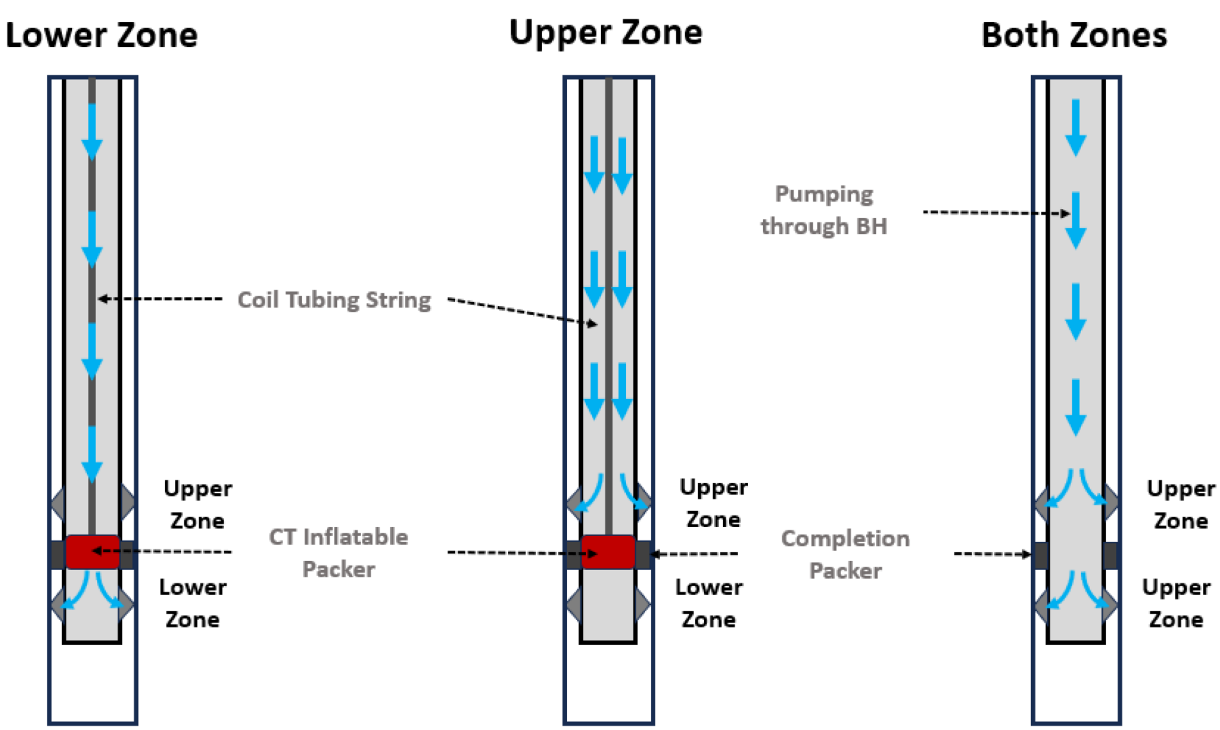

Figure 3—Illustration of zonal injection strategies using coiled tubing inflatable packer or Bullheading strategy.

From above figures, CT is run and placed strategically isolating the targeted zone from the rest. From Fig. 3 if the lower zone is targeted then the chemical would be pumped inside the coil and in case of the upper zone, chemical is pumped via coiled tubing – Tubing annulus ensuring total isolation from the upper and lower zones. To enhance tool reliability under extreme conditions, ac active cooling mechanism was implemented by continuously pumping water @ 0.5~1 barrel per minute (BPM) into the non-treated zone during the operation to reduce the harsh temperature of <500 °F to packer working limit (350 °F) . This cooling method of the elements allows the packer to sustain its deflation and ensures no communication between the upper and lower zone. In cases where both upper and lower zones required treatment, a bullheading operation was performed without any mechanical barriers to distribute the chemical across both intervals simultaneously.

Execution

The project was executed in three phases from 2020 to 2024, with over 100 wells treated by the end of the campaign. Following the success of Phase 1 in 2020—which was discussed in detail in [3] (SPE-207254-MS)—the project moved into Phase 2 in 2021, during which 85 wells were successfully treated between 2021 and 2022. Additionally, two initial pilot wells were executed in Egypt to evaluate cross-regional applicability. The final phase involved more than 18 wells, treated using a combination of coiled tubing unit (CTU) operations and bullheading techniques. Each well had its own treatment design, tailored to the reservoir condition, completion constraints, and injectivity characteristics.

Upon CTU rig-up, two primary runs were conducted. The first was a drift run utilizing a spin-cut nozzle to ensure clear access to the target depth. This step not only confirmed mechanical clearance for the inflatable packer but also allows cooling of the wellbore through continuous water circulation during run hole (RIH). After assembling the subsequent Bottom Hole Assembly (BHA) with the spin-cut nozzle and completing the required pressure testing, the coiled tubing was injected into the well at a controlled speed of 15 ft/min while maintaining a water pumping rate of 1 BPM inside the CT string. Upon passing any potential well jewelry, the run-in speed was increased to 50 ft/min, and pumping was sustained. Pull tests were performed every 1,000 ft to monitor string weight and validate it against simulation data. Once the coiled tubing tagged the known target depth (TD)—indicated by a noticeable weight slack—the pumping rate was raised to 2.5 BPM to clean out the completion and lift any debris. If an obstruction was encountered prior to reaching TD, efforts to penetrate were made by continuing to pump at 2.5 BPM until the recorded TD was reached. Subsequently, injector and reel depths were Correlated against TD, and the coiled tubing was pulled out of hole (POOH) to the packer setting depth. The string was flagged at the designated setting point to eliminate any depth discrepancies during the second run.

During the second run, the inflatable packer was deployed while maintaining a minimum circulation rate of 1 BPM in the CT–tubing annulus to protect the packer from high-temperature exposure. Once the flag depth was reached, the CT was run 5 ft deeper and then picked up to ensure the string was in tension. Accounting for the Bottom Hole Assembly (BHA) length difference between the two runs, the CT was pulled up by 4 ft, reflecting the known offset. After bleeding off pressure in the CT string, a ball was dropped to initiate packer activation following the pre-defined sequence. Once the ball was dropped, pumping was resumed at 1 BPM to drift the ball down the string. Based on the known CT string volume inside the reel, the operator pumped up to twice the internal volume until the ball reached the top of the injector—confirmed by the absence of fluid turbulence noise inside the reel. The ball was then left to descend under gravity for 10 to 15 minutes. After this waiting period, the pump was re-engaged at a minimum rate to shear the packer activation pins. Successful setting was indicated by a distinct pressure spike reaching approximately 2,500 psi, followed by a sudden drop, as illustrated in Fig. 4.

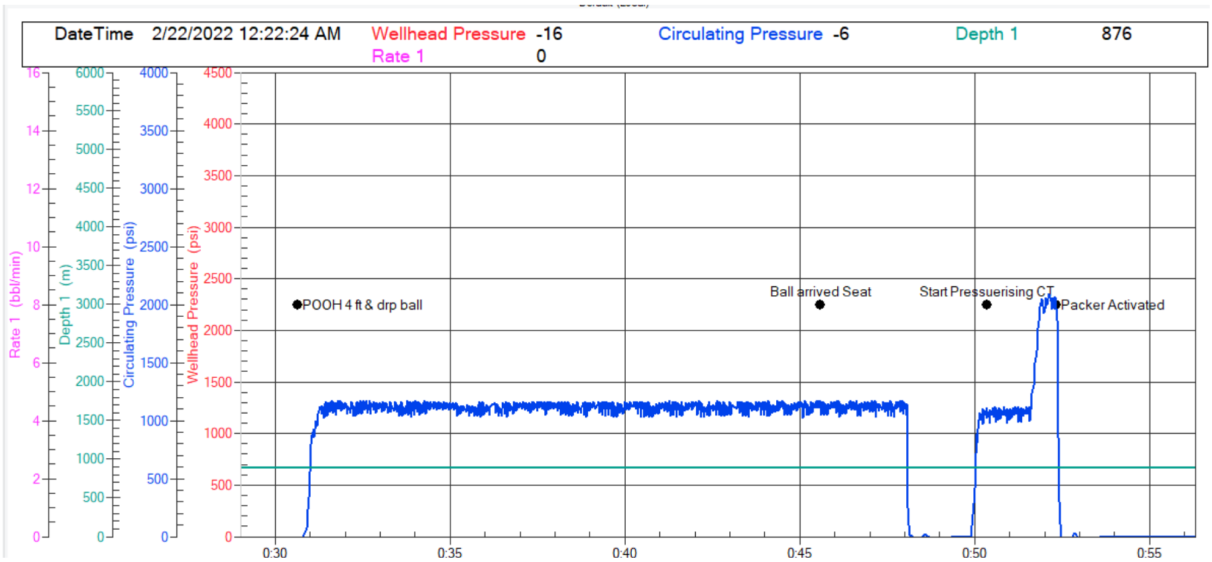

Figure 4—Operation pressure chart illustrating packer activation sequence.

Following zonal isolation, over 100,000 gallons of conformance control chemical were pumped at the maximum permissible rate based on injectivity. Cooling water was continuously injected into the non-treated zone at 1 BPM to preserve packer integrity. For upper zone treatments, chemical was pumped through the CT–tubing annulus, and for lower zones, directly via CT. In dual-zone cases, a bullheading strategy was used. The average operation lasted around 15 hours per well. Fig. 5 shows a representative case where the chemical was injected at 3 BPM while simultaneously injecting water through CT at 0.5–1 BPM. Pressure and rate charts indicated a steady increase in WHP and stable circulation pressures, confirming effective zone isolation and chemical reaction buildup.

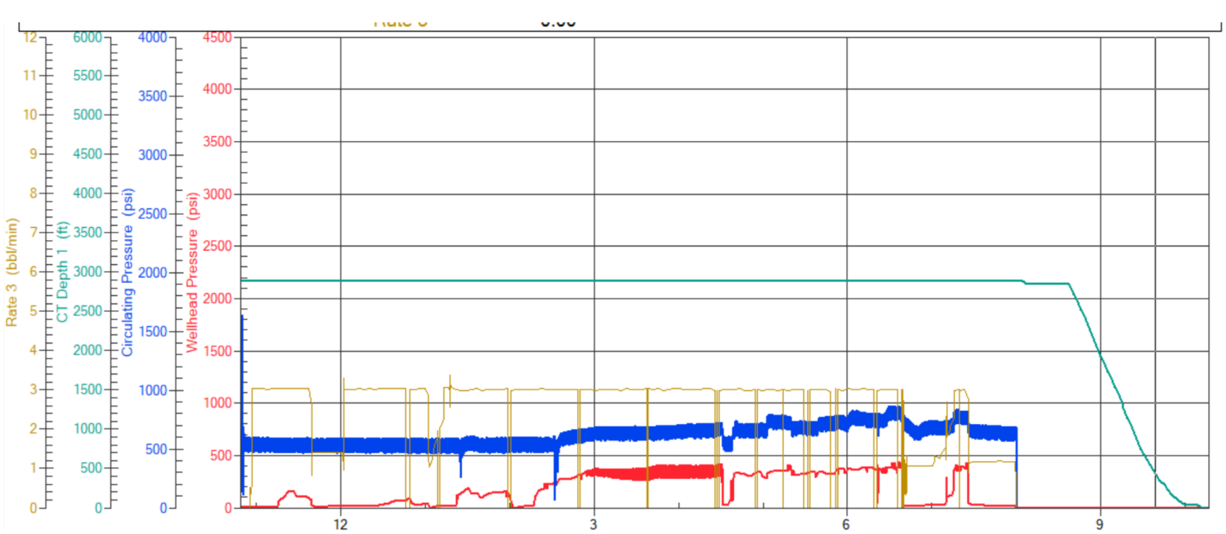

Figure 5—Surface pressure and rate chart during coiled tubing intervention

As the treatment progressed, a gradual increase in wellhead pressure (red) was observed—indicating a reduction in pore throat diameter due to the advancing chemical reaction. Meanwhile, the circulation pressure (blue) remained stable, confirming that the inflatable packer successfully isolated the treated zone from the non-treated intervals.

Evaluation

Based on data collected from the Data Acquisition System, Fig. 6, demonstrates a progressive increase in wellhead pressure along the treatment stages, as shown on the primary Y-axis. To stay within the maximum allowable annulus surface pressure (MAASP), the injection rate (secondary Y-axis) was gradually reduced. This pressure response is a strong indicator of chemical buildup within the formation, confirming the plugging effect of the conformance solution as it penetrates and narrows the pore throats in the targeted zone.

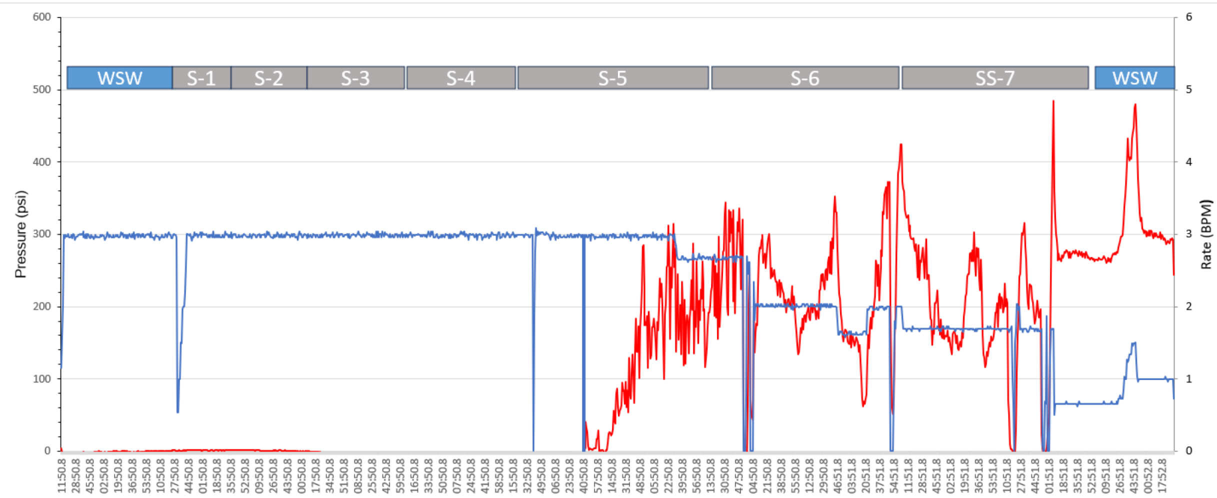

Figure 6—Surface pressure and rate trend across treatment stages.

Following (CTU) deployment, a spinner log was conducted to evaluate the steam injection conformance profile across the wellbore. As illustrated in Figure 7, the pre-conformance profile indicates a highly unbalanced steam distribution, with 89% of the injection entering the upper zone and only 11% reaching the lower zone. This imbalance reflects the presence of a dominant thief zone in the upper interval, leading to inefficient sweep and underutilization of the lower zone.

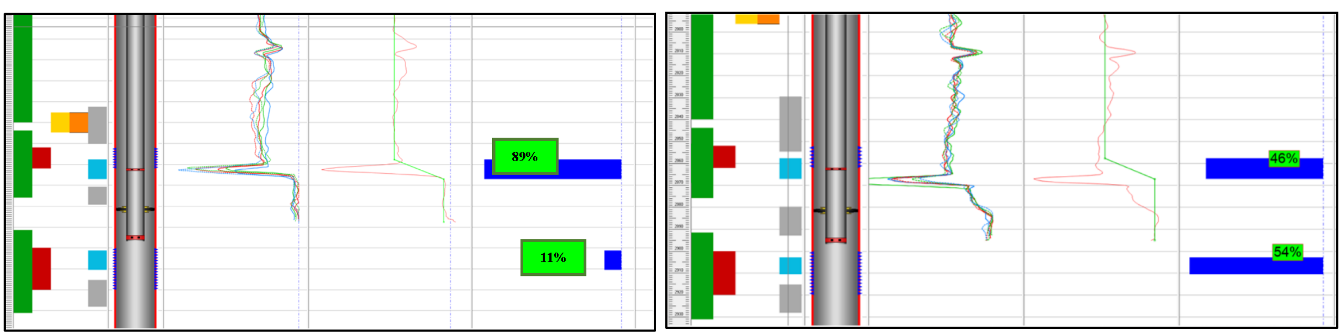

Figure 7—Spinner log comparison showing injectivity profile pre (left)- and post (right)-conformance treatment.

After the conformance control treatment was applied using conformance control chemical, the post-job injectivity profile demonstrated a significant improvement in steam distribution. The spinner log revealed a near-even split, with 46% of the injection directed to the upper zone and 54% to the lower zone. This shift confirms successful flow diversion and improved sweep efficiency, resulting in better utilization of the previously under-swept lower zone. The redistribution of injectivity enhances oil recovery by engaging more of the reservoir volume and reducing steam override.

The enhanced steam distribution achieved through the conformance treatment translated directly into operational and economic benefits. With improved sweep efficiency, less steam was required to lift the same volume of oil, thereby significantly reducing lifting costs. Simultaneously, by successfully engaging previously unswept reservoir zones, the oil recovery efficiency was notably increased. A direct environmental benefit was also realized: with reduced steam consumption, with the associated CO_2 emissions reduction, contributing to the project's sustainability goals. Furthermore, the conformance control chemicals used in this campaign were locally manufactured through small and medium enterprise (SME) partnerships, reinforcing the in-country value (ICV) strategy. This not only supported the local economy but also fostered technical expertise and knowledge transfer, positioning the solution for potential application in similar thermal EOR environments beyond Oman's borders.

Conclusion

The deployment of a thermally stable conformance control strategy in the Mukhaizna heavy oil field successfully addressed the long-standing challenge of poor steam sweep efficiency caused by thief zones. Utilizing coiled tubing and inflatable packers, the operation demonstrated an innovative, field-ready solution for delivering chemical treatments selectively and effectively under high-temperature (>500°F) and high-H_2S conditions.

A significant improvement in steam injection conformance was achieved, with injectivity profiles shifting from a severely skewed 89%/11% distribution to a balanced 46%/54% between the upper and lower zones, respectively. This re-distribution improved reservoir contact, enhanced oil recovery, and minimized steam override. Surface and downhole monitoring confirmed successful zone isolation and sustained chemical placement, validated by stable circulation pressures and rising wellhead pressures—indicators of pore throat plugging.

The operational design included active cooling to safeguard packer integrity, precise depth correlation techniques, and adaptable deployment methods (including bullheading) for dual-zone treatment. The project's execution across more than 100 wells over four years proved the strategy's robustness and scalability.

Economically, the treatment led to reduced lifting costs by minimizing unnecessary steam injection. Environmentally, CO_2 emissions were reduced by up to 25%, aligning with sustainability objectives. Additionally, the use of locally manufactured chemicals through SME partnerships bolstered Oman's in-country value (ICV) framework and opened pathways for technology export.

This work offers a replicable model for thermal EOR conformance in similar reservoirs globally and underscores the synergy between engineering ingenuity, chemical science, and operational discipline in enhancing field performance.

Nomenclature

Term / Abbreviation	Definition
API	American Petroleum Institute gravity – measure of oil density
BHA	Bottom Hole Assembly
BHT	Bottom Hole Temperature
BPM	Barrels Per Minute
CT	Coiled Tubing
CTU	Coiled Tubing Unit
EOR	Enhanced Oil Recovery
HPHT	High Pressure High Temperature
H_2S	Hydrogen Sulfide
ICV	In-Country Value
MAASP	Maximum Allowable Annulus Surface Pressure
OD	Outer Diameter

POOH	Pull Out of Hole
RIH	Run In Hole
SME	Small and Medium Enterprises
SOR	Steam to Oil Ratio
TD	Target Depth
WHP	Wellhead Pressure

Reference

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